DAY 2 | PRE-READ

Talking to Computers and Organising Things

A light read before Day 2. About 20 minutes, no code to memorise. This includes the self-study topics: Python setup and syntax basics.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Tomorrow you write your first Python, then learn to group instructions, organise information and save it so it is still there the next day. This note gives you the big ideas in everyday language, so the class feels familiar instead of new.

1. What is a program?

A program is a list of instructions that a computer follows exactly, in order. Think of a recipe: what you need, the steps, and what to do if something changes ("if the batter is too thick, add milk"). A computer never guesses. It does exactly what you write.

Key idea: computers are not smart, they are obedient. Clear instructions give clear results.

2. Self-study: setup and syntax basics

We use **Google Colab**, which runs Python in your browser, and **VS Code** on your laptop from Day 3. Please do these before class:

- Sign in to a Google account and open colab.research.google.com once.
- Create a notebook, type print("Hello, Python!") and press Shift + Enter.
- Lines starting with # are comments. Python ignores them.
- Indentation matters: use 4 spaces to group lines under if, for, while and def.
- Python is case sensitive: name, Name and NAME are three different things.
- Follow shared/Laptop_Setup_Guide.pdf to install Python 3.12 or newer and VS Code.

3. Words you will hear

Word	Everyday meaning
Variable	A labelled jar that holds a piece of information
Condition	A yes or no question that decides what happens next
Loop	Doing the same step again until the job is done
Function	A kitchen appliance: inputs go in, a result comes out

Word	Everyday meaning
File	A notebook that keeps data after the program ends

4. Four ways to organise things

Real life	Python name	What makes it special
A shopping list	list	Ordered; you can add, remove or change items
Your date of birth	tuple	Fixed; it does not change
Unique stamps in a collection	set	No duplicates allowed
A phone contact	dictionary	Find information by a label, called a key

5. Why save to files?

When a program ends, everything in memory disappears, like a whiteboard that gets wiped. Saving to a file is like writing it in a notebook. **CSV** is a simple table that opens in Excel. **JSON** stores labelled, nested information and is used by almost every app and API.

6. Think about these before class

- Write down the steps for making a cup of tea. How many "if" decisions did you include?
- Write down the details you store for one contact on your phone. Which are labels and which are values?
- A customer calls a telecom helpdesk. What details would you write on the ticket?

Kaizen (改善): continuous improvement, one small step at a time